EXPERIMENT 1

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

Data Science Trends, Roles, Data Types and Encryption

Aim: Visualize job trends and roles, identify data types, and encrypt/decrypt text.

Python Program

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# 1(a) Job trend
import pandas as pd
import matplotlib.pyplot as plt
df = pd.DataFrame({'Year':range(2010,2021),'Jobs':[150,300,450,600,800,1200,1600,2100,2700,3400,4200]})
plt.plot(df.Year,df.Jobs,marker='o'); plt.show()

# 1(b) Role distribution
roles=['Analyst','Engineer','Scientist','ML Engineer','Business Analyst']
plt.bar(roles,[300,500,450,200,150]); plt.xticks(rotation=30); plt.show()

# 1(c) Data types
print(pd.DataFrame({'ID':[1,2],'Name':['A','B']}))
print("Unstructured: text, image, video")
print({'ID':1,'Name':'A','Details':{'Age':20}})

# 1(d) Encryption
from cryptography.fernet import Fernet
f=Fernet(Fernet.generate_key())
msg=b"Hello Data Science"
token=f.encrypt(msg)
print(token)
print(f.decrypt(token))
```

Result

The plots show job trends and role counts. Examples of three data types are displayed, and the encrypted text is decrypted.

Note: For CSV programs, keep the named CSV file in the same folder and check that column names match.